

Biodiversity and Ecosystem Functioning

Today's Agenda:

- Quiz
- BEF: Unanswered Questions

Unanswered Questions in BEF

Your book highlight some important areas where work is still needed:

- Effects across multiple trophic levels
- Species loss vs introduction
- Extinction is global, functions are local
- The invasion paradox
- The importance of diversity in natural systems
- Complicated relationship between diversity and productivity

Unanswered Questions in BEF

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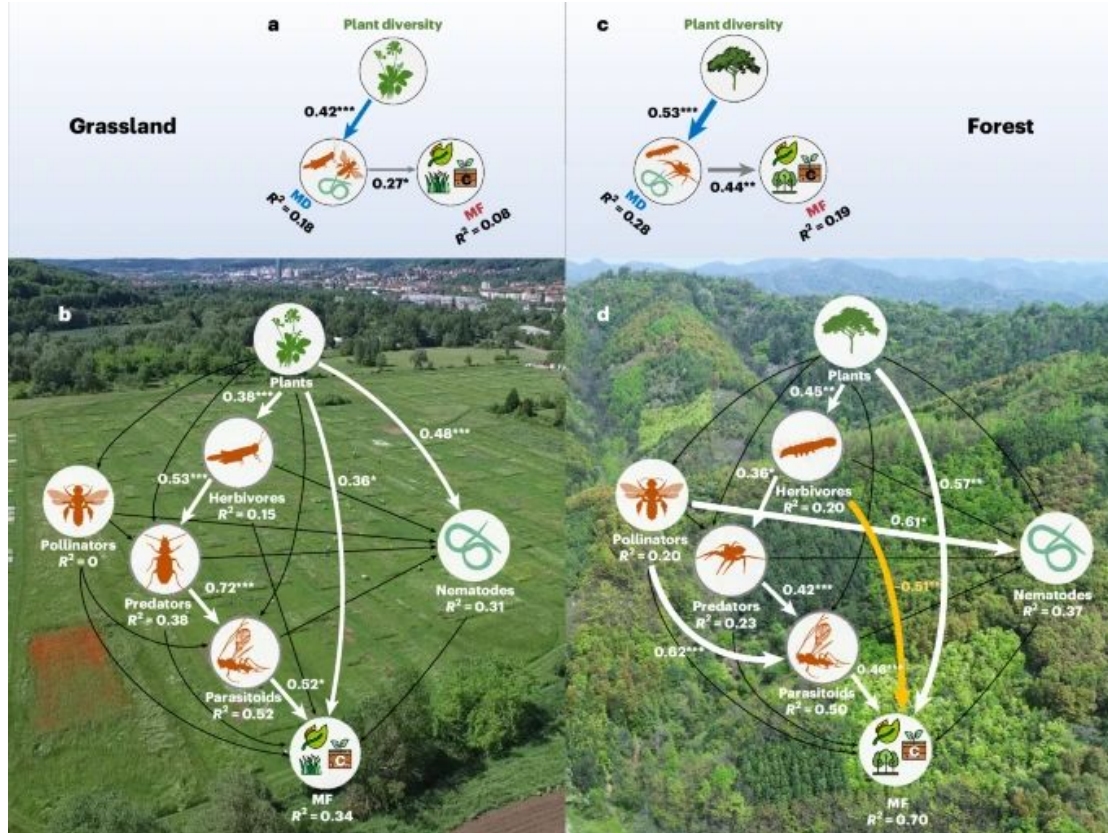
- Effects across multiple trophic levels
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Effects across trophic levels

Most BEF experiments only manipulate plant diversity

While many monitor diversity of other groups, they don't often manipulate it

Effects across trophic levels



nature ecology & evolution

Article <https://doi.org/10.1038/s41559-024-02517-2>

Plant diversity enhances ecosystem multifunctionality via multitrophic diversity

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 Accepted: 24 July 2024
 Published online: 29 August 2024
 Check for updates

Yi Li¹, Andreas Schüldt², Anne Ebeling², Nico Eisenhauer^{1,3}, Yuanyuan Huang^{4,5}, Georg Albert², Cynthia Albracht⁶, Angelos Amyntas⁷, Michael Bonkowski⁸, Helge Bruelheide^{9,10}, Maximilian Bräcker¹¹, Douglas Chesters¹², Jun Chen¹³, Yunnan Chen^{10,11}, Jing-Ting Chen^{10,11}, Marcel Giobanu¹², Xianglu Deng¹², Felix Fornoff¹⁴, Gerd Gleixner¹⁵, Liangdong Guo^{13,16}, Peng-Fei Guo¹⁷, Anna Heintz-Buschart¹⁸, Alexandra-Maria Klein^{14,19}, Markus Lange¹⁰, Shan Li¹, Qi Li¹⁰, Yingbin Li¹⁰, Arong Luo¹⁰, Sebastian T. Meyer²⁰, Goddert von Oheimb²¹, Gemma Rutten²², Thomas Scholten²³, Marcel D. Solbach²⁴, Michael Staab²⁵, Ming-Qiang Wang^{26,27}, Naili Zhang²⁸, Chao-Dong Zhu^{29,30}, Bernhard Schmid³¹, Keping Ma³¹ & Xiaojuan Liu^{11,32}

This paper we discussed yesterday is a step in the right direction!

Why is it important?

We're losing trophic complexity
E.g., top predators, large herbivores

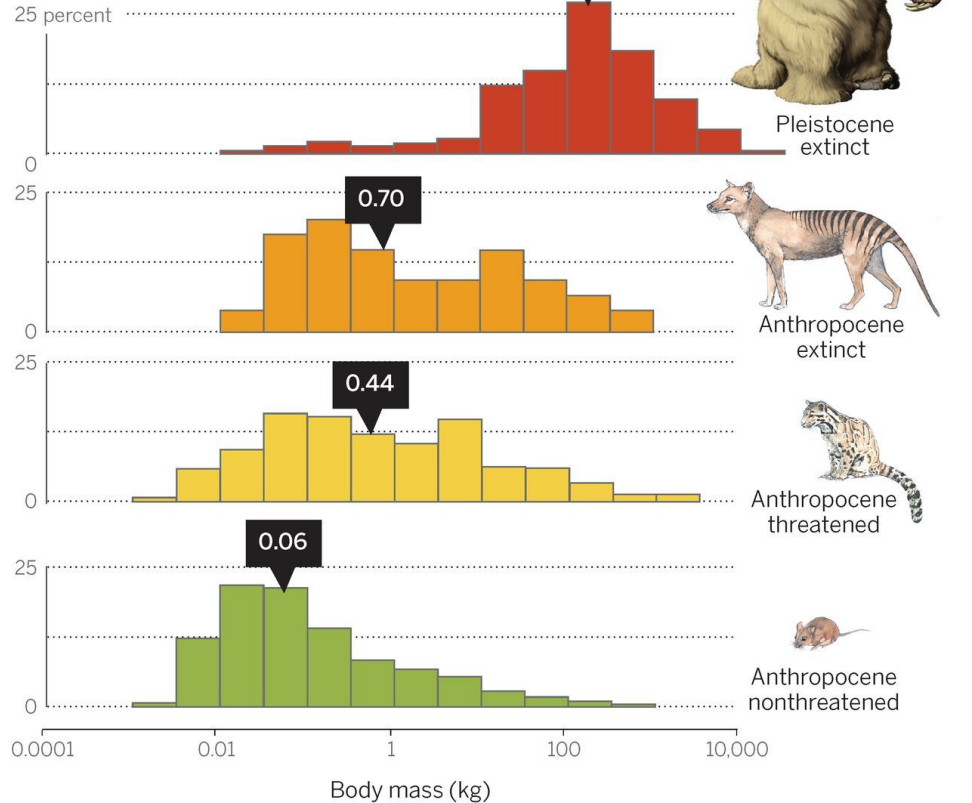
REVIEW

Defaunaion in the Anthropocene

Rodolfo Dirzo,^{1*} Hillary S. Young,² Mauro Galetti,³ Gerardo Ceballos,⁴
Nick J. B. Isaac,⁵ Ben Collen⁶

Size-differential defaunaion

Frequency of extinction (median value highlighted)

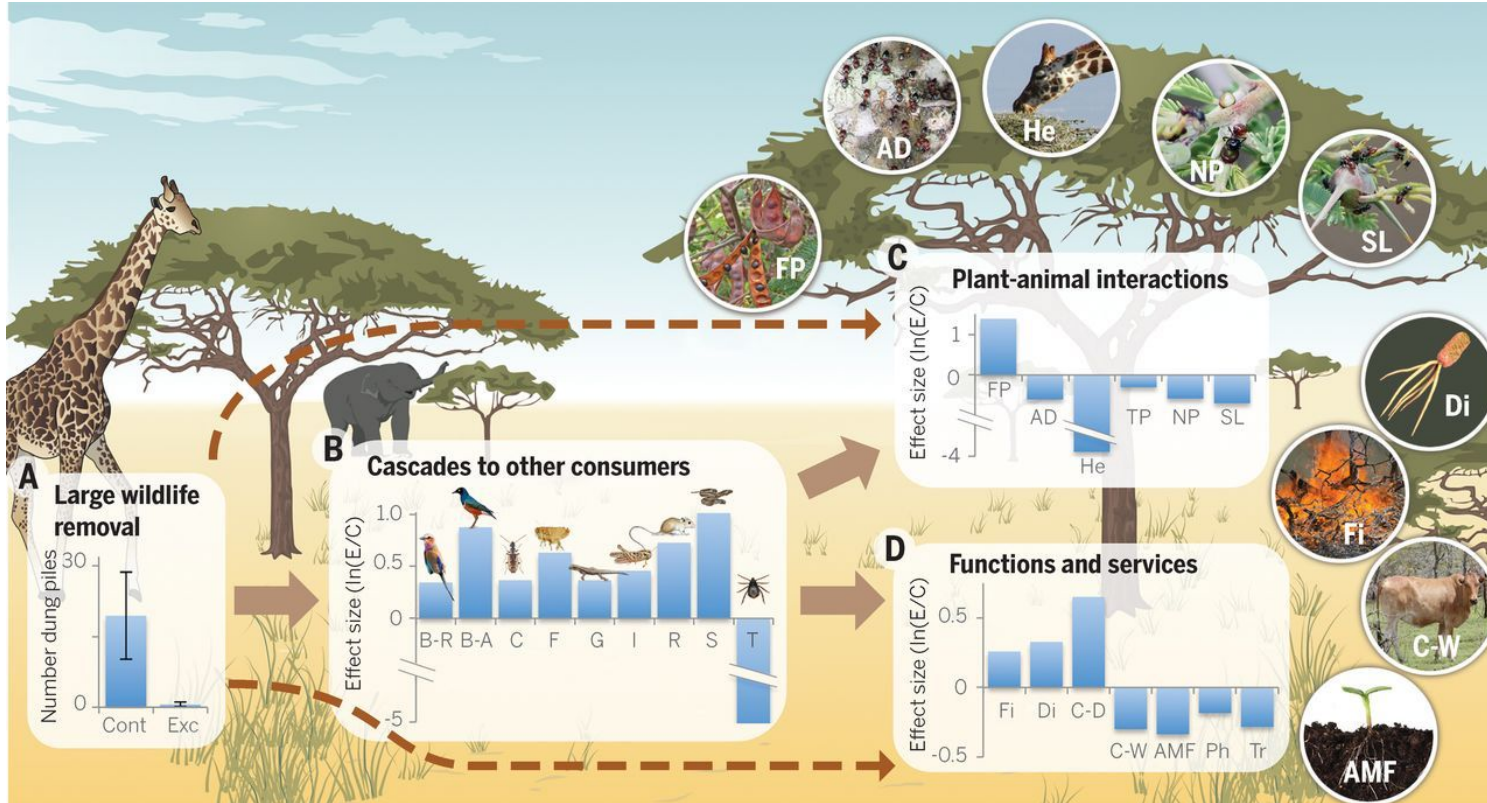


Why is it important?

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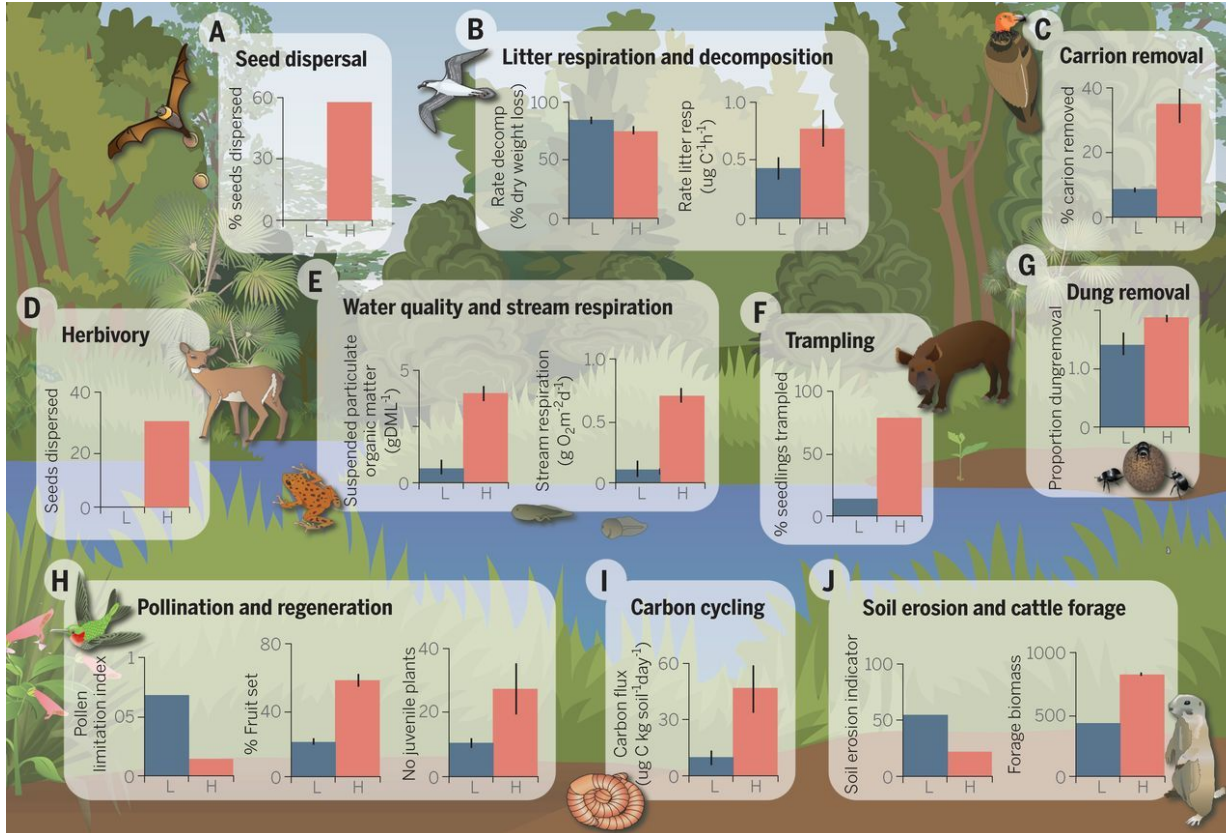
Removing large wildlife can have large impacts!

Why is it important?

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Removing consumers in general can also have impacts, even the smaller consumers!

Effects across trophic levels

- We know the higher trophic levels matter for functioning
- Unclear how *diversity* in higher trophic levels impacts things
- Also unclear how diversity interacts across trophic levels
 - (e.g., do herbivore diversity effects depend on plant diversity?)

Effects across trophic levels

- We know the higher trophic levels matter for functioning
- Unclear how *diversity* in higher trophic levels impacts things
- Also unclear how diversity interacts across trophic levels
 - (e.g., do herbivore diversity effects depend on plant diversity?)
- Algal or microbial systems in the lab are probably a good place to test!

Unanswered Questions in BEF

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Species loss vs introduction

Most BEF experiments focus on assembling species

Individuals of selected species are added to an unoccupied environment

Species loss vs introduction

Most BEF experiments focus on assembling species

Individuals of selected species are added to an unoccupied environment

Adding species to an empty area == removing species from an occupied area?

Testing BEF with removal

Island experiment in Sweden

- 30 islands, 14 plots each (n = 420 total)
- Removed 0 to 3 functional groups (tree, shrub, moss)
- Removed 0 to 3 species of shrub
- Measured 15 ecosystem functions

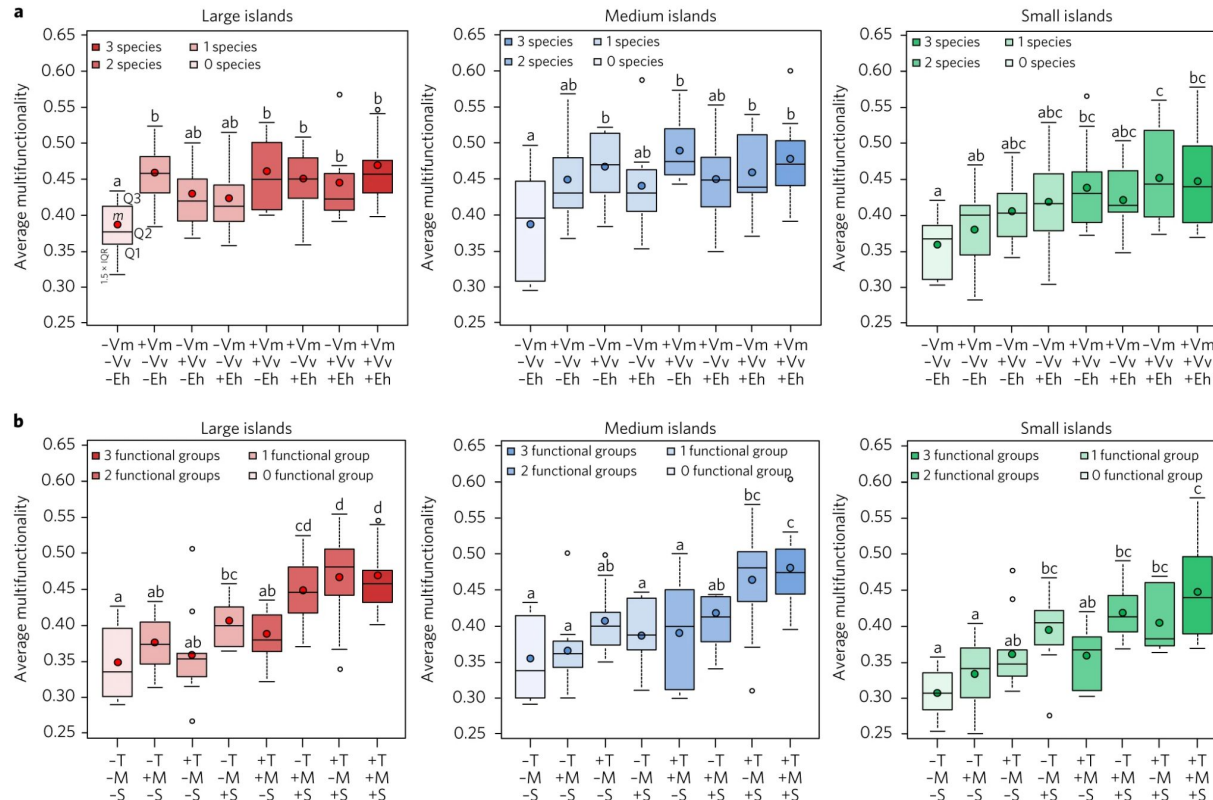
Consistent effects of biodiversity loss on multifunctionality across contrasting ecosystems

Nicolas Fanin^{1,2*}, Michael J. Gundale¹, Mark Farrell¹, Marcel Ciobanu¹, Jeff A. Baldock¹, Marie-Charlotte Nilsson¹, Paul Kardol¹ and David A. Wardle^{1,5}

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BEF and removal

Need for experiments that use both addition and removal

- Relatively fast systems probably most amenable (e.g. algae)
- Some experiments suggest that order of arrival matters

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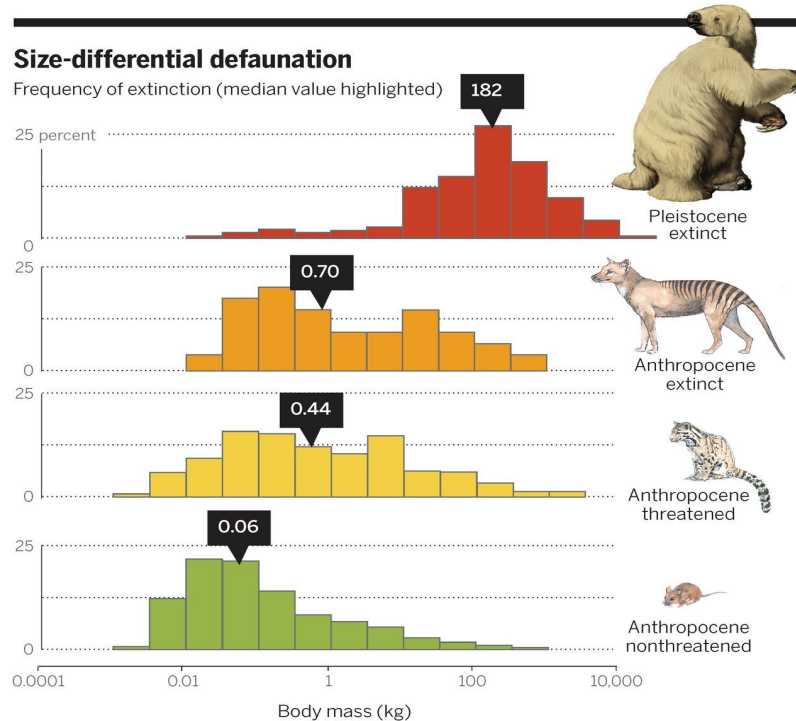
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Extinction is global, functions are local

Strong evidence that humans are causing lots of extinctions

Extinction is global, functions are local

Strong evidence that humans are causing lots of extinctions



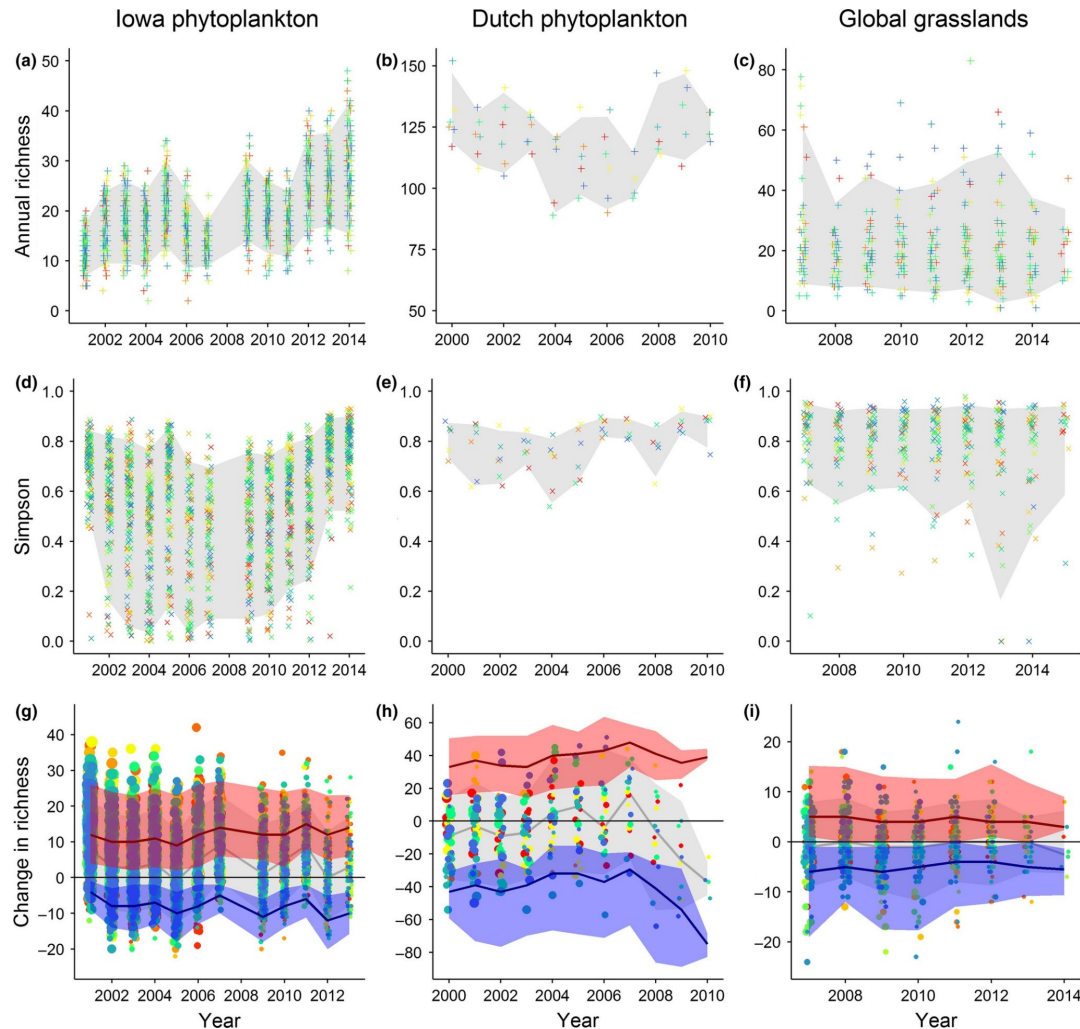
REVIEW

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Net local change

But species introductions can
balance out local extirpations



Received: 15 March 2017 | Accepted: 13 June 2017
DOI: 10.1111/1365-2664.12959

RESEARCH ARTICLE*

Journal of Applied Ecology

**Biodiversity change is uncoupled from species richness trends:
Consequences for conservation and monitoring**

Helmut Hillebrand^{1,2} | Bernd Blasius^{2,3} | Elizabeth T. Borer⁴ | Jonathan M. Chase^{5,6} |
John A. Downing⁷ | Britas Klemens Eriksson⁸ | Christopher T. Filstrup⁷ | W. Stanley
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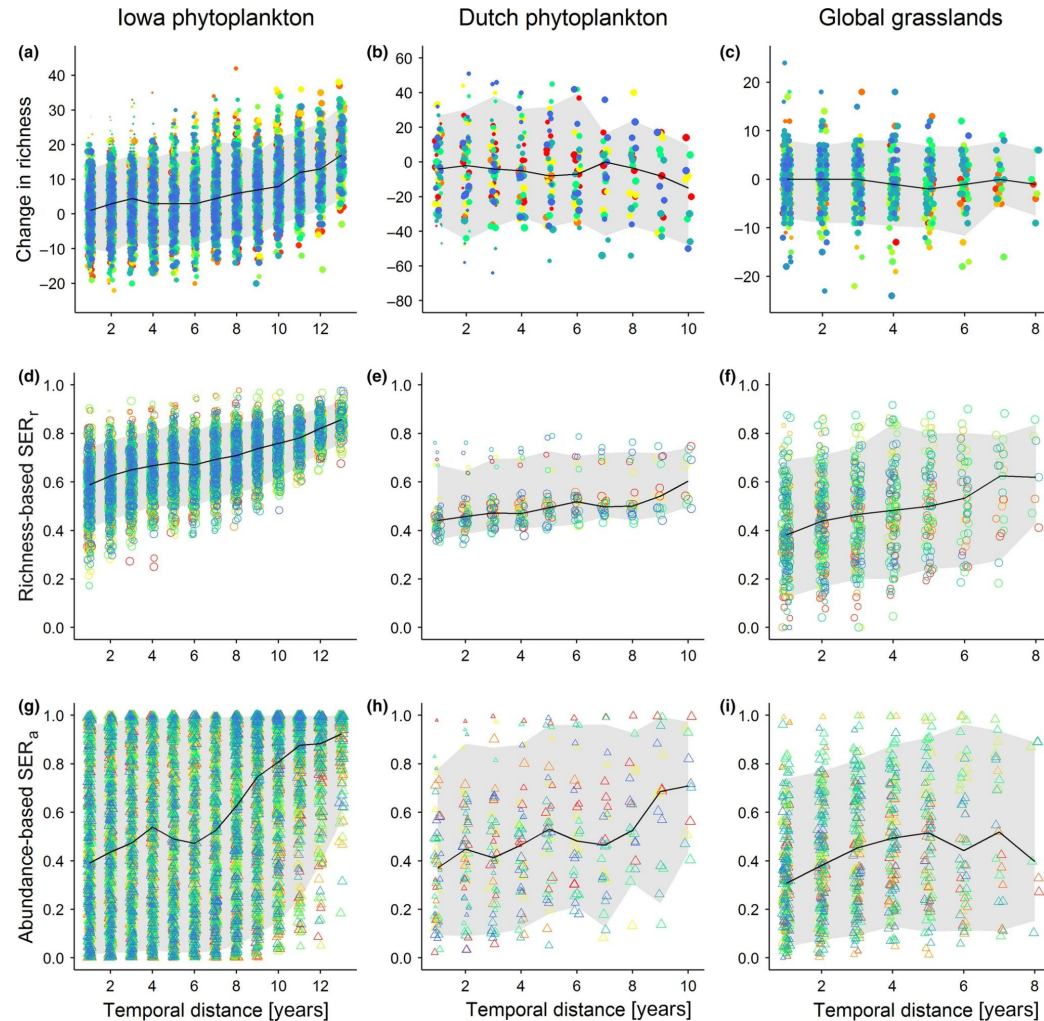
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But species introductions can
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SER = Species Exchange Ratio

$$SER_r = \frac{S_{imm} + S_{ext}}{S_{tot}}$$

$$SER_a = \frac{\sum_i (p_i - p'_i)^2}{\sum p_i^2 + \sum p'_i^2 - \sum p_i p'_i}$$



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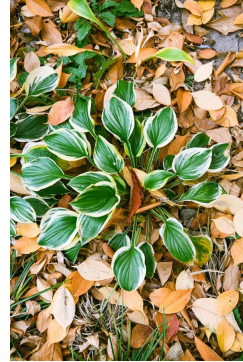
Need to go beyond species richness

Species richness only gets us so far

- Richness can stay constant despite major changes in species
- Moving to a trait-based approach may help



VS



Unanswered Questions in BEF

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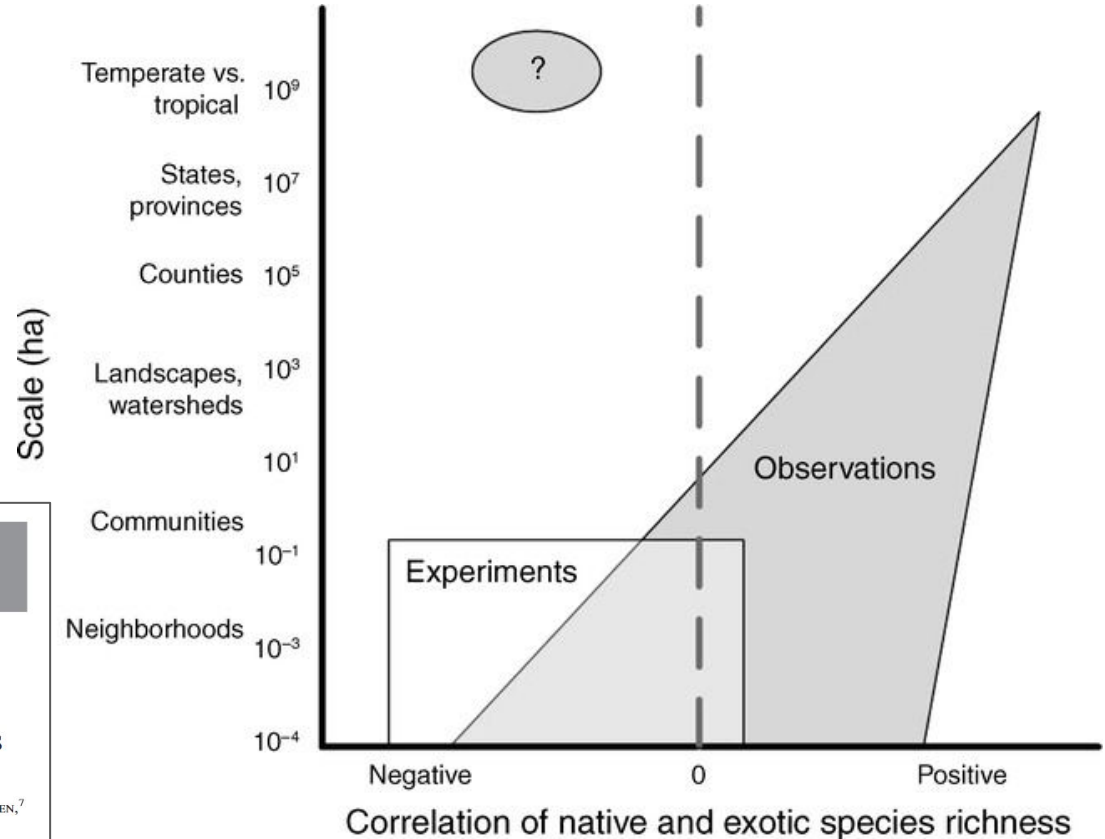
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The Invasion Paradox



CONCEPTS & SYNTHESIS

EMPHASIZING NEW IDEAS TO STIMULATE RESEARCH IN ECOLOGY

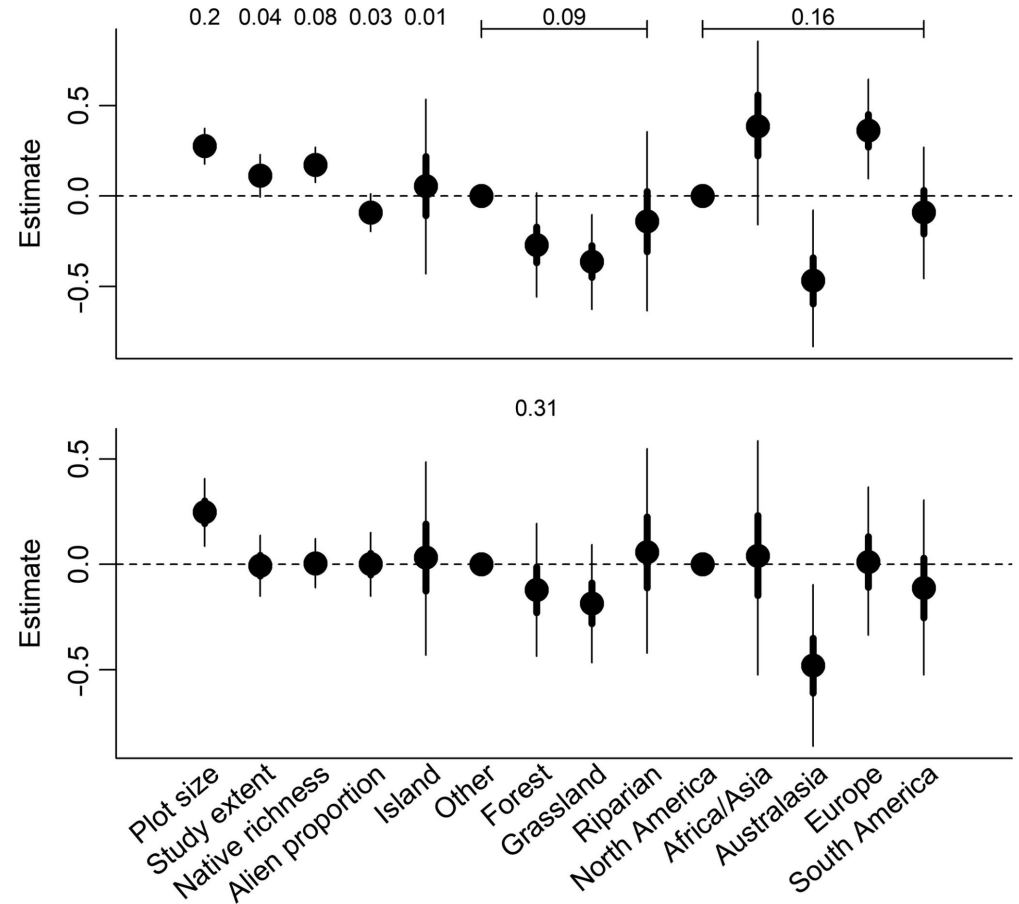
Ecology, 88(1), 2007, pp. 3–17
© 2007 by the Ecological Society of America

THE INVASION PARADOX: RECONCILING PATTERN AND PROCESS IN SPECIES INVASIONS

J. D. FRIDLEY,^{1,10} J. J. STACHOWICZ,² S. NAEEM,³ D. F. SAX,⁴ E. W. SEABLOOM,⁵ M. D. SMITH,⁶ T. J. STOHLGREN,⁷
D. TILMAN,⁸ AND B. VON HOLLE⁹

The Invasion Paradox

Top panel: Separate models
Bottom panel: Single model



ECOLOGY LETTERS

Ecology Letters, (2019) 22: 1038–1046

doi: 10.1111/ele.13261

REVIEW AND SYNTHESIS

Resolving the invasion paradox: pervasive scale and study dependence in the native–alien species richness relationship

Abstract

The degree to which plant communities are vulnerable to invasion by alien species has often been assessed using the relationship between native and alien plant species richness (NAR). Variation in the direction and strength of the NAR tends to be negative for small plot sizes and study

Federico Tomasello,^{1*} Richard P. Duncan^{2,3} and Philip E. Hulme²

The Invasion Paradox

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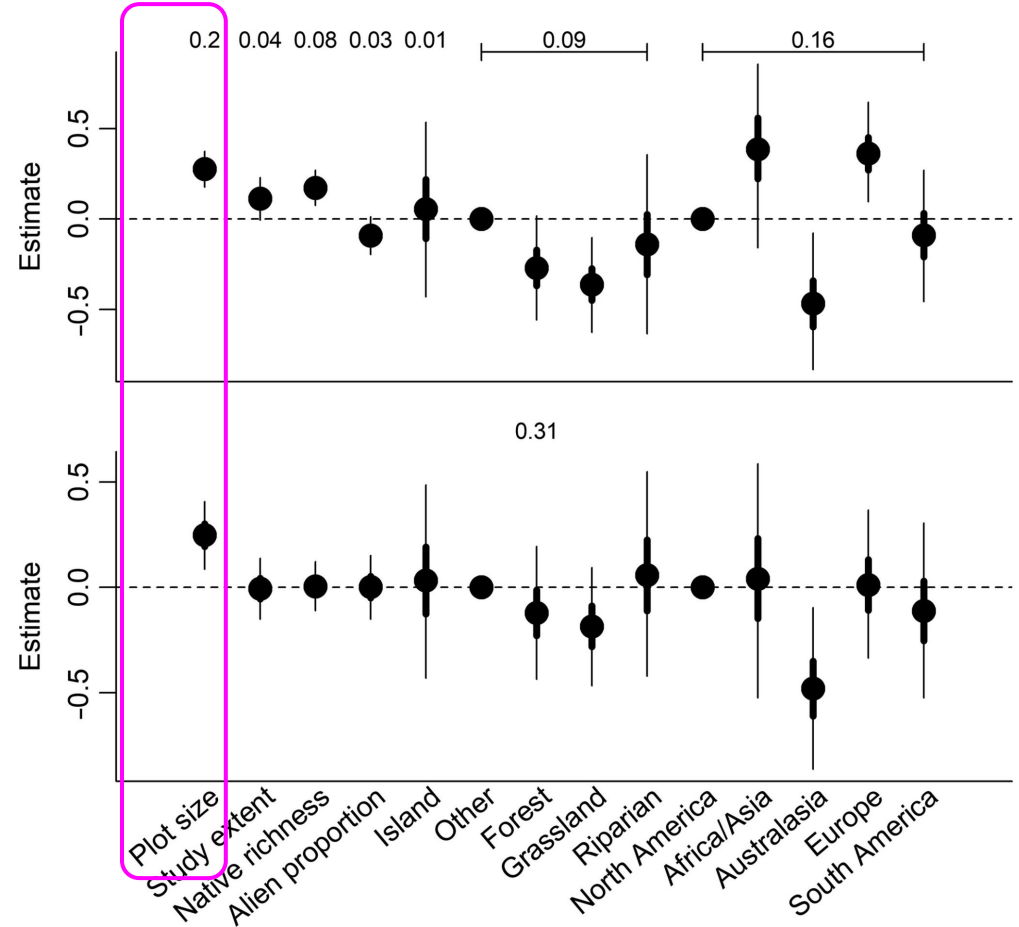
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Federico Tomasetto,^{1*} Richard P. Duncan^{2,3} and Philip E. Hulme²



The Invasion Paradox

Why?

- Same factors that drive native species richness drive introduced species richness?
- Experiments often relatively homogeneous, as opposed to more natural systems
- More work needs to be done

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BEF in natural systems

- Much of what we know is from experimental systems
- How important are these effects in natural systems?

BEF in natural systems

Biodiversity and ecosystem functioning in naturally assembled communities

Fons van der Plas*

Systematic Botany and Functional Biodiversity, Institute of Biology, Leipzig University, Johannisallee 21-23, 04103 Leipzig, Germany

					Observed relationships		
Category of function	Ecosystem function	Ecosystem type	Focal group	Theory	Experiments	Non-experimental studies <i>N</i>	
Biomass	Tree biomass stock	Temperate forests	Trees	✓ _{a,b,c}	✓ _m	38	✓
	Tree biomass production	Temperate forests	Trees	✓ _{a,b,c}		61	✓
	Tree biomass stock	Tropical forests	Trees	✓ _{a,b,c}	✓ _m	44	⇒
	Tree biomass production	Tropical forests	Trees	✓ _{a,b,c}		17	⇒
	Plant biomass*	Grasslands	Plants	✓ _{a,b,c}	✓ _n	102	✓
	Plant biomass*	Aquatic systems	Plants	✓ _{a,b,c}	✓ _n	21	✓
	Consumer biomass	All	Consumers	✓ _{a,b,c}		24	✓
Decomposition	Decomposition	All	Plants	✗ _d	✓ _o	33	✓
	Decomposition	All	Decomposers	✓ _d	✓ _p	20	⇒
Soil carbon storage	Soil organic carbon stock	All	Plants		✓ _o	35	⇒
Biomass stability	Plant biomass stability	All	Plants	✓ _{e,f,g}	✓ _{q,r}	27	✓
	Consumer biomass stability	All	Consumers	✓ _{e,f,g}		13	✓
Pathogen / herbivore damage	Overall pathogen damage	All	Hosts		✗ _o	17	⇒
	Damage by specialist pathogen	All	Hosts	✗ _h		18	✗
	Herbivore damage	All	Plants	✗ _{h,i}	⇒ _o	45	⇒
	Herbivore damage	All	Herbivores	✓ _{j,k}		10	⇒
	Herbivore damage	All	Predators	✗ _k	⇒ _o	11	⇒
Pollination	Fruit or seed set	All	Plants			8	⇒
	Fruit or seed set	All	Pollinators	✓ _l		36	✓
Ecosystem multi- functionality	Ecosystem multifunctionality	All	All		✓ _s	111	✓
	Ecosystem multifunctionality	Temperate forests**	All			16	✓
	Ecosystem multifunctionality	Tropical forests**	All			17	⇒
	Ecosystem multifunctionality	Grasslands**	All			55	⇒

*In grasslands and aquatic systems, distinguishing between biomass stocks and production is challenging, hence I pooled biomass stock and production BEF relationships in grasslands.

**These analyses rely on a subset of the data (and are therefore not independent) of the analysis on ecosystem multifunctionality across all ecosystem types.

BEF in natural systems

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	Damage by specialist pathogen	All	Hosts	✗ _h		18 
	Herbivore damage	All	Plants	✗ _{h,i}	✗ _o	45 
	Herbivore damage	All	Herbivores	✓ _{j,k}		10 
	Herbivore damage	All	Predators	✗ _k	✓ _o	11 
Pollination	Fruit or seed set	All	Plants			8 
Ecosystem multi- functionality	Fruit or seed set	All	Pollinators	✓ _l		36 
	Ecosystem multifunctionality	All	All		✓ _s	111 
	Ecosystem multifunctionality	Temperate forests**	All			16 
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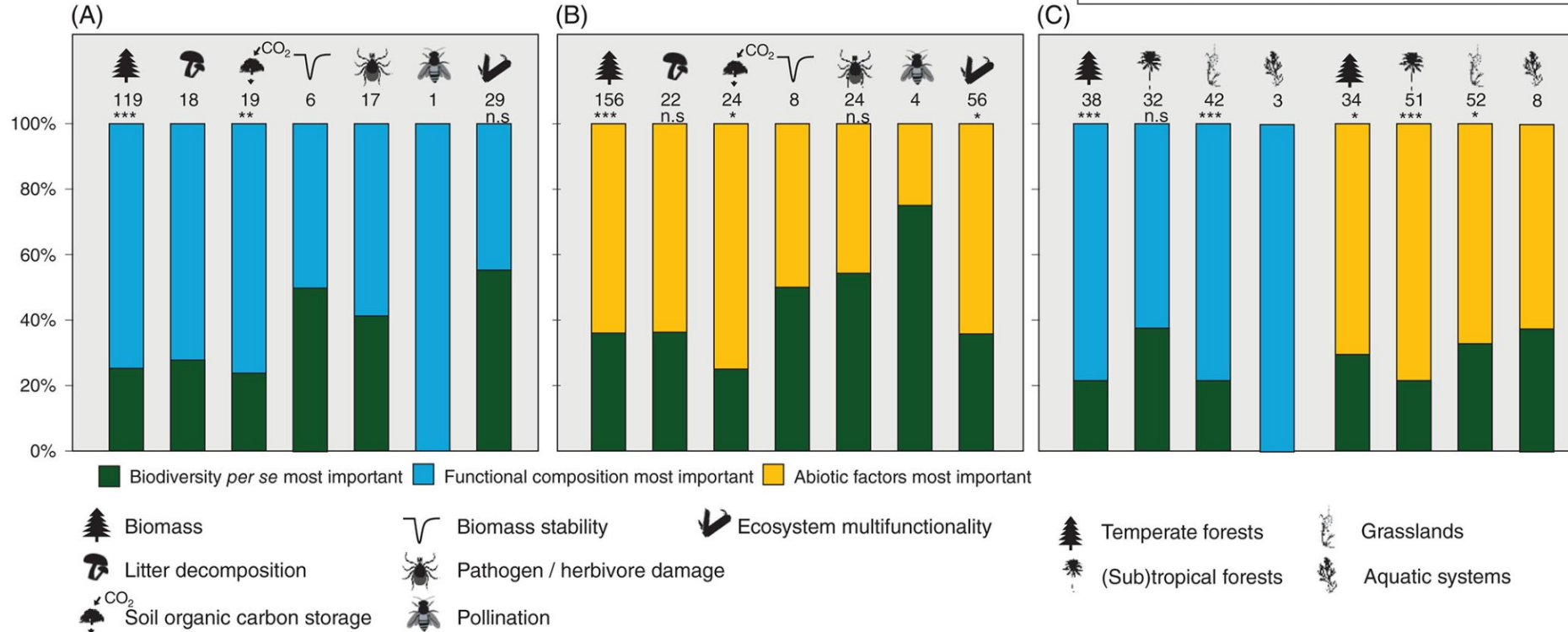
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Biodiversity and ecosystem functioning in naturally assembled communities

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BEF in natural systems



BEF in natural vs experimental systems

- Some relationships seems consistent, others differ
- Richness isn't usually the main factor - functional composition matters more

Unanswered Questions in BEF

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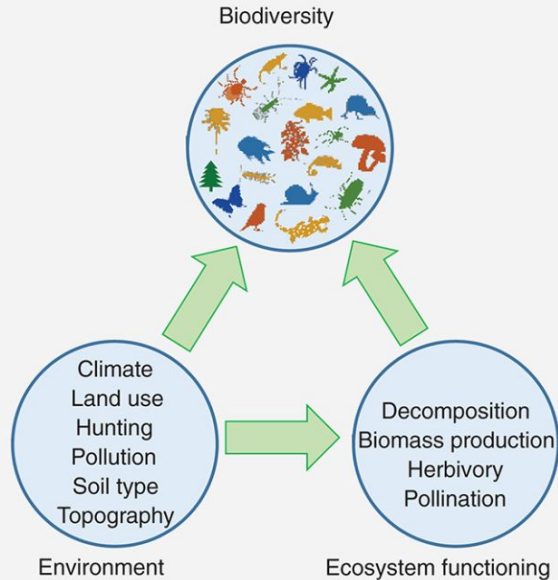
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BEF: It's complicated

(A)

Old paradigm: Environmental variation drives biodiversity and ecosystem functioning

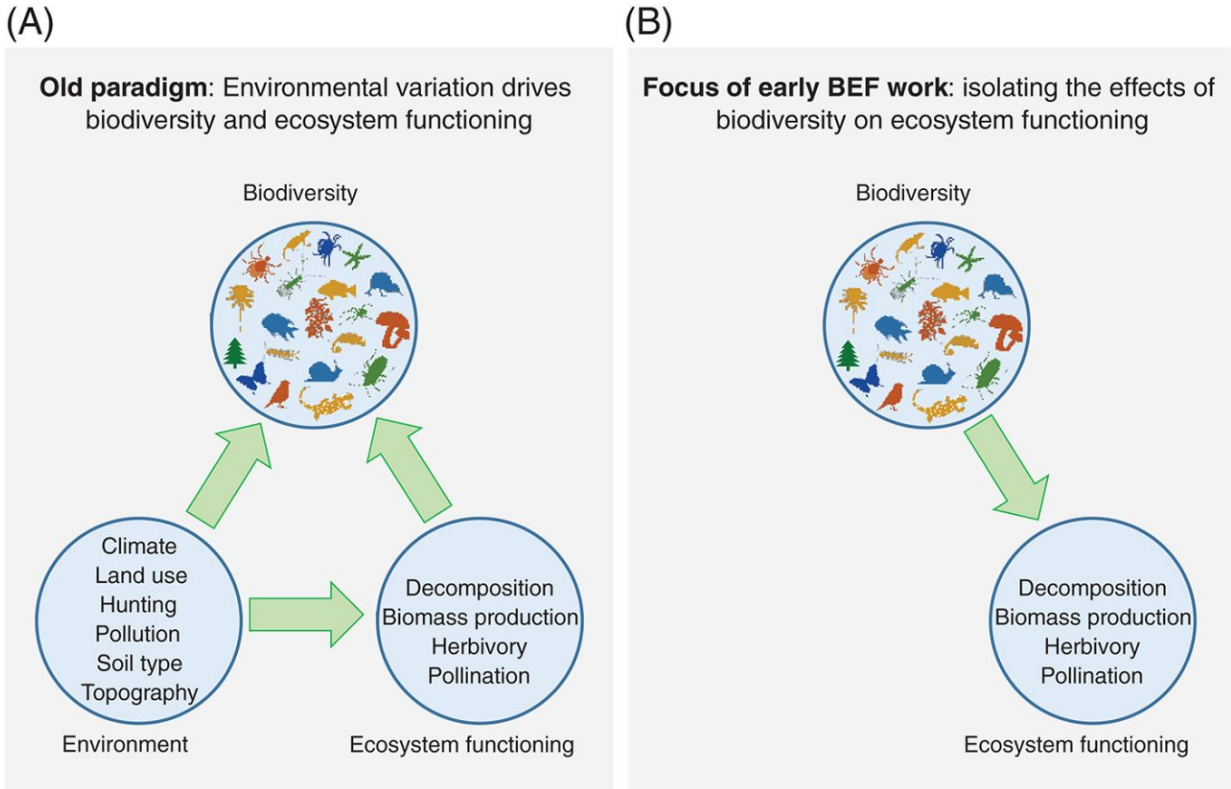


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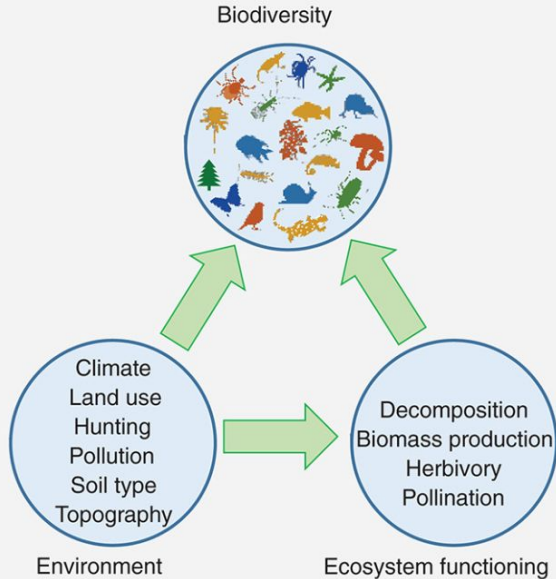
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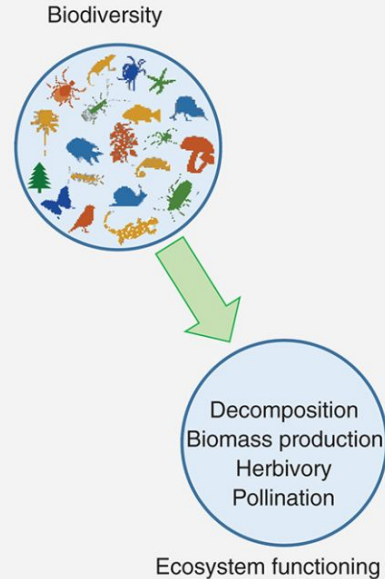
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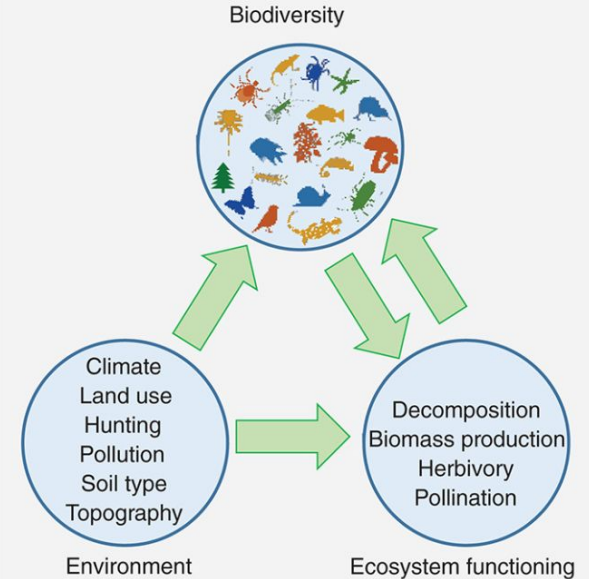
(B)

Focus of early BEF work: isolating the effects of biodiversity on ecosystem functioning



(C)

New paradigm: biodiversity and environmental variation jointly drive ecosystem functioning

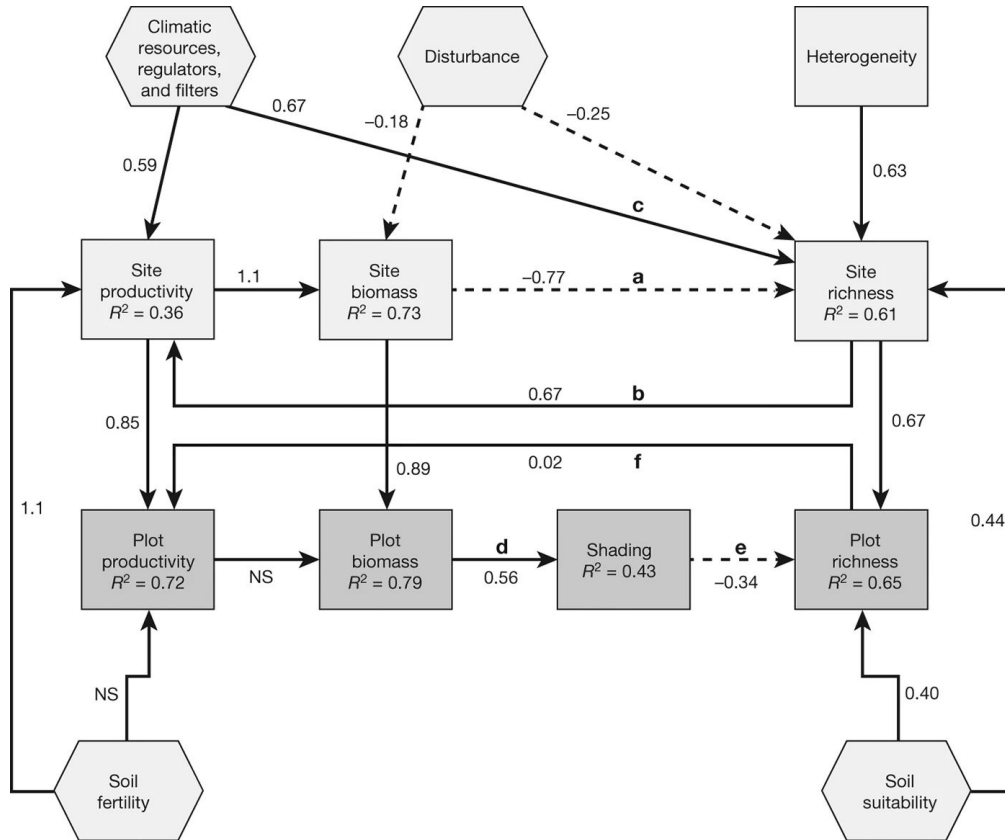


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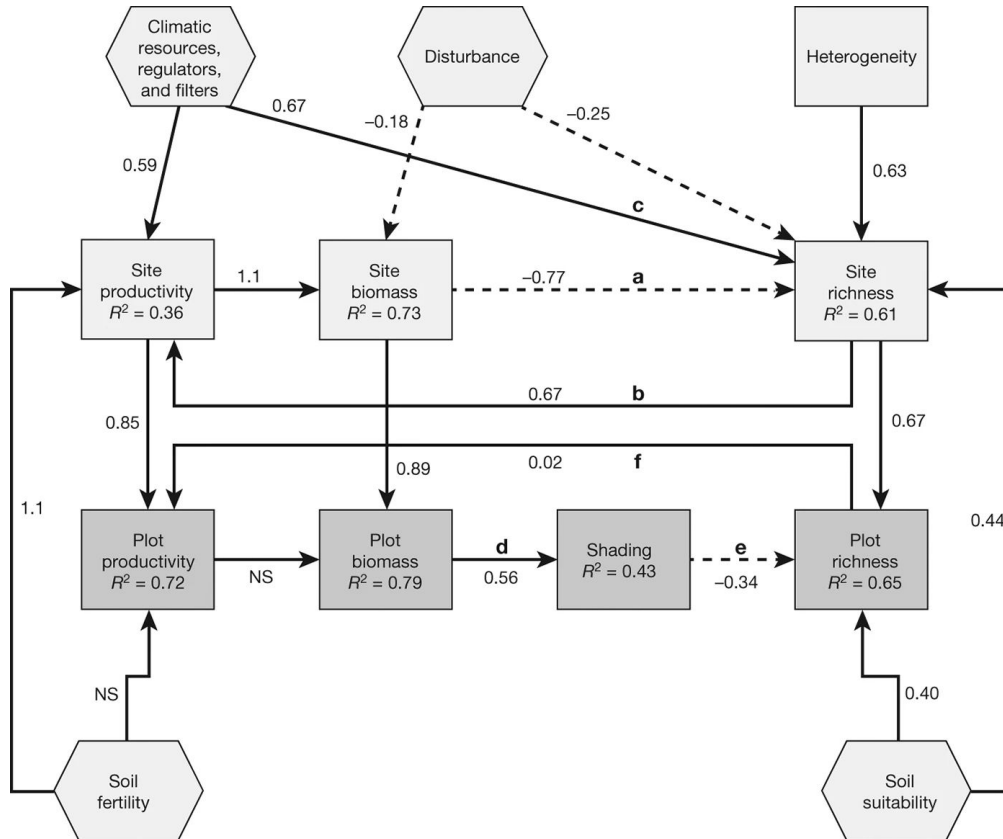
BEF: It's complicated



Integrative modelling reveals mechanisms linking productivity and plant species richness

James B. Grace¹, T. Michael Anderson², Eric W. Seabloom³, Elizabeth T. Borer³, Peter B. Adler⁴, W. Stanley Harpole^{5,6,7}, Yann Hautier⁸, Helmut Hillebrand⁹, Eric M. Lind³, Meelis Pärtel¹⁰, Jonathan D. Bakker¹¹, Yvonne M. Buckley¹², Michael J. Crawley¹³, Ellen I. Damschen¹⁴, Kendi F. Davies¹⁵, Philip A. Fay¹⁶, Jennifer Firn¹⁷, Daniel S. Gruner¹⁸, Andy Hector¹⁹, Johannes M. H. Knops²⁰, Andrew S. MacDougall²¹, Brett A. Melbourne¹⁵, John W. Morgan²², John L. Orrock¹⁴, Suzanne M. Prober²³ & Melinda D. Smith²⁴

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Complicated relationships might also explain differences between experiments and nature!

Or variation among systems!

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Next class: Density Dependence!